

BUSINESS + TECHNICAL SUMMARY

WISER DOM Hybrid Solver

A portable classical default with safeguarded quantum experimentation

Executive decision

Adopt the validated classical hierarchy: use **polished greedy** for routine planning and trigger **exact large-neighborhood search (LNS)** when inventory, capacity, or penalty conflicts justify additional runtime. Keep the bounded sampler-assisted path as a controlled research option behind exact recourse and independent validation.

On the common 100-assignment-group subset, greedy improves objective capture from 74.96% to 76.82% and case fill from 78.29% to 79.91% in 0.58 seconds. Exact quantity polish reaches 76.83% capture in 2.05 seconds. Exact LNS retains the strongest nominal frontier in 13.22 seconds. A time-limited full MILP returns a lower incumbent with a nonzero gap, demonstrating the value of preserving the hierarchy’s strong incumbent.

Business problem and measures

Distributed order management selects a distribution center (DC) and ship date while orders compete for shared time-phased inventory and capacity. Orders sharing a load must move together; an alternate DC must improve service; and no recommendation may exceed cumulative available-to-promise (ATP) inventory or documented capacity.

The optimized objective is fulfilled merchandise value minus unmet-demand penalty minus shipping cost. **Objective capture** normalizes that value by total requested merchandise value, avoiding disclosure of commercial totals. **Case fill** is fulfilled cases divided by requested cases. **Reassignment** means moving an order away from its default DC.

Solution hierarchy

Layer	What it does	Recommended use
Default	Keeps planned routing and allocates feasible quantities	Business baseline
Greedy	Chooses whole-load alternatives while updating resources	Fast incumbent
Polished greedy	Fixes routes and exactly re-optimizes quantities and penalties	Routine certified plan
Exact LNS	Reopens bounded groups of interacting decisions	Quality escalation
Full MILP	Solves all selected decisions and reports incumbent, bound, and gap	Small-scope comparison
Hybrid sampler	Proposes bounded local assignments; exact models retain control	Research

The hybrid QUBO contains only local assignment choices. Exact enumeration, random sampling, simulated annealing, local QAOA, or hardware-executed QAOA can propose a bitstring. The workflow repairs one-hot violations, exactly re-optimizes fulfillment, validates the complete plan, and rejects invalid or non-improving results. QAOA is the proposal method; an IBM **backend** is a processor/execution target, not a solver.

Audited evidence

The final study contains **516 aggregate rows across 14 experiment families**. Of these, 513 return plans and all 513 pass the independent validator with zero recorded demand, integrality, inventory, and capacity residuals. Three frozen-routing controls are correctly proven infeasible at 60%, 65%, and 70% inventory reductions.

Common 100-group method	Objective capture	Case fill	Runtime
Default routing	74.96%	78.29%	0.24 s
Greedy	76.82%	79.91%	0.58 s
Polished greedy	76.83%	79.91%	2.05 s
Exact LNS	76.83%	79.91%	13.22 s
Full MILP, time-limited	76.80%	79.91%	1.68 s
Hybrid simulated annealing	76.83%	79.91%	55.74 s

All scalable methods reach 372 real assignment groups and 750 orders. Median full-scope runtime is 1.31 seconds for greedy, 4.82 for polished greedy, 26.14 for exact LNS, and 101.17 for hybrid simulated annealing. On generated instances, greedy and exact LNS reach 100,000 orders in 39.16 and 276.71 seconds; hybrid reaches 20,000 orders while the local QUBO remains 32 variables.

The inventory frontier shows why adaptive routing matters. Frozen nominal routing is feasible through a 55% inventory reduction and then fails. Adaptive methods remain feasible at 70%; exact LNS achieves 72.74% objective capture and 76.99% case fill, compared with 72.16% and 76.56% for greedy.

Quantum and hardware result

A generated coordination control requires a joint move that greedy misses. Exact LNS, full MILP, simulated annealing, local QAOA, and hardware-derived QAOA proposals all recover the 197.2 synthetic-unit improvement after recourse, increasing fill from 21.49% to 30.58%.

The final IBM study contains 18 jobs at 8,192 shots each on a 16-variable control. At $p=1$, median raw one-hot feasibility is 65.34%, exact-optimum hit rate is 0.6836%, transpiled depth is 43, and the circuit uses 40 two-qubit gates. At $p=2$, these become 10.36%, 0.0610%, depth 448, and 373 gates. Shallow QAOA therefore retains the stronger signal. All final plans remain protected by exact recourse and validation. End-to-end runtime is reported on a log scale with individual-job dots because one long queue wait would otherwise hide the typical 42-54 second medians.

Deployment and next decisions

The default stack is license-free and runs on Windows, macOS, and Linux with Python 3.10-3.12. SciPy/HiGHS is the portable exact engine; native HiGHS, SCIP, Gurobi, GPU scoring, and IBM Runtime are opt-in.

Before a pilot, confirm authorized alternative DCs, working calendars, throughput limits, customer rules, and approval thresholds. Run polished greedy over rolling historical windows; trigger exact LNS for material conflicts; review validator diagnostics and planner acceptance; and compare realized service with default routing. Restrict hardware reruns to approved generated or bounded neighborhoods with exact and uniform-feasible controls.

The submission delivers a usable scalable solver now and a disciplined test bed for future proposal methods without making operational quality depend on a license, GPU, or quantum account.